



Multiple modes of AFM reveal distinct mechanical properties for dystrophin and utrophin not manifest by small fragments

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Duchenne muscular dystrophy (DMD) is a lethal muscle disease caused by the absence of the protein dystrophin. Dystrophin is hypothesized to work as a molecular shock absorber that limits myofiber membrane damage when undergoing reversible unfolding upon muscle stretching and contraction. Here, we report the mechanical characterization of single full-length dystrophin (Dys) molecules using two operational modes of atomic force microscopy; constant speed and constant force as well as Monte Carlo simulations. Furthermore, we have compared Dys with large fragments encoding the N-terminus through spectrin repeat 10 (DysN-R10), the C-terminal retinal isoform of dystrophin (Dp260), and full-length utrophin (Utr). Our comprehensive data reveal that Dys, DysN-R10, and Dp260, all show a uniform, brittle unfolding behavior, whereas Utr demonstrates more complex unfolding dominated by a stiffening spring behavior. These fundamentally different mechanical behaviors *in vitro* suggest different *in vivo* functions for Dys and Utr with implications for the potential efficacy of Utr upregulation to substitute for Dys deficiency in DMD.

dystrophin | utrophin | force spectroscopy | constant speed and constant force experiments | mechanical properties

Duchenne muscular dystrophy (DMD) is a lethal muscle disease that affects 1 in 5,000 boys born in the United States (1) characterized by progressive muscle degeneration and weakness due to loss of the protein dystrophin (2). Dystrophin is a 427 kDa protein expressed primarily at the muscle cell membrane, or sarcolemma, in striated muscle tissue and is a crucial component of the dystrophin–glycoprotein complex (DGC) (3). It is hypothesized that dystrophin acts as a molecular shock absorber to stabilize the sarcolemma and protect muscle against mechanical forces during muscle stretching and contraction (4–10). Dystrophin deficiency increases sarcolemmal fragility ultimately leading to myofiber death (11).

The functional domain architecture of dystrophin provides insights into how its absence leads to disease (9). At the N-terminus, an actin-binding domain anchors dystrophin to actin filaments beneath the sarcolemma (Fig. 1A). This is followed by a central rod domain composed of 24 spectrin-like triple-helix repeats interspersed with four flexible hinge regions, forming a mechanical linkage between the actin-binding domain and the dystrophin–glycoprotein complex (DGC)-binding domains. DGC-binding is mediated by globular domains located C-terminal to the spectrin-like repeats. It is hypothesized that the central spectrin-like repeats unfold in response to mechanical stress, allowing dystrophin to absorb mechanical energy and protect the sarcolemma from damage. However, the mechanical properties of full-length dystrophin have yet to be fully characterized, and it remains unclear whether they support its proposed role as a molecular shock absorber.

Upregulation and subsequent overexpression of a naturally homologous protein, utrophin, is proposed as a potential DMD therapy with the goal of substituting utrophin for dystrophin at the sarcolemma (12). Utrophin (Fig. 1A) is abundantly expressed during fetal development, and like dystrophin, forms a complex with actin filaments and the DGC at the sarcolemma but is later replaced by dystrophin after birth (13). In adult skeletal muscle, utrophin primarily localizes to the myotendinous and neuromuscular junctions (14). Utrophin overexpression is shown to compensate for dystrophin deficiency in mdx mice (15) and could do so in humans as well. How utrophin's mechanical properties compare with dystrophin is not known.

In this study, we characterized the mechanical properties of full-length dystrophin (Dys) using atomic force microscopy (AFM), and compared it with full-length utrophin (Utr) and large dystrophin fragments encoding the N-terminus through repeat 10 (DysN-R10) and Dp260 encoding ~R11–C-terminus (Fig. 1A). We performed single molecule force spectroscopy (SMFS) using both constant speed and constant force modes of

Significance

Duchenne muscular dystrophy (DMD) is a lethal muscle wasting disorder caused by loss of the protein dystrophin. Utrophin is a fetal homologue of dystrophin that is under active investigation as a dystrophin replacement therapy for DMD. However, it is unknown if utrophin can substitute for dystrophin from a mechanical perspective. Here, we directly compared the mechanical properties of full-length dystrophin with utrophin using atomic force microscopy through two operational modes; constant speed and constant force. Our data reveal distinct mechanical behaviors for dystrophin and utrophin that suggest their functions are different in a physiological context.

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The authors declare no competing interest.

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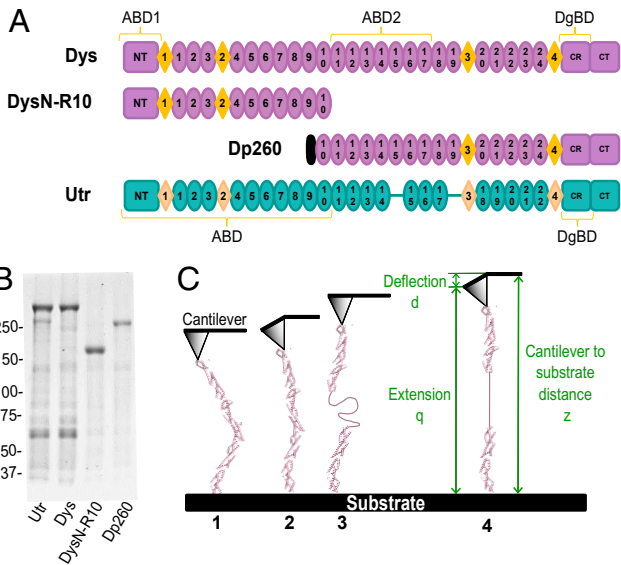


Fig. 1. Single molecule force spectroscopy of dystrophin and utrophin. (A) Domain structure of Dys, DysN-R10, Dp260, and Utr. Ovals, spectrin-like repeats (SLRs); diamonds, hinge domains; NT, N terminus; CR, C terminus; DgBD, dystroglycan-binding domain; ABD1 & 2, actin-binding domains; and DgBD, dystroglycan binding domain. (B) Coomassie blue-stained SDS polyacrylamide gel with purified proteins. (C) Illustration of the SMFS experiment.

AFM. The statistics of force magnitude data from constant speed mode and the time to unravel statistics of the constant force mode were independently consistent. We analyzed these data globally using the Dudko-Hummer-Szabo (DHS) model (16), which serves as a cornerstone for relating statistics from the two operational modes and extracting kinetic information to model domain unfolding via Monte Carlo simulations. Such consistency ensures data reliability (17, 18). Our data demonstrate that Dys, DysN-R10, and Dp260, all exhibit brittle unfolding behavior while the mechanical unfolding of Utr is distinct and more complex, marked by a stiffening spring behavior. Our direct in vitro comparison of Dys and Utr suggests different in vivo functions. Notably, although the early unfolding behaviors of Dys and Utr are similar, significant differences arise during the later unfolding events. We hypothesize these differences will impact their interchangeability as therapeutics.

Results

Constant Speed and Constant Force Experiments Yield Consistent Unfolding Transition Rates for Dystrophin. The proteins analyzed in this study are depicted in Fig. 1A and their quality assessed on Coomassie blue-stained SDS polyacrylamide gels (Fig. 1B). Dys and Utr preparations each contained 10% of a ~60 kDa band (Fig. 1B), which was identified by mass spectrometry to be asparagine-tRNA ligase derived from the insect cell host. We do not believe that the presence of this minor contaminant impacted subsequent analyses by AFM because Dys and Utr presented with significantly different unfolding profiles (Fig. 4) even though both were contaminated to similar extents while the unfolding profiles for DysN-R10 and Dp260 matched closely that of Dys (Fig. 4) even though preparations of both lacked the contaminating protein (Fig. 1B). We also assessed the potential contribution of asparagine-tRNA ligase to our data and found it represents less than 5% of traces, confirming that it does not meaningfully affect our results (see SI Appendix for details). AFM operational modes of constant speed and constant force

were utilized for SMFS experiments. One portion of the protein randomly attaches to the substrate and the remainder is left free to attach to the force probe, as illustrated by state 1 in Fig. 1C. The cantilever to substrate distance, z , is retracted at a constant rate in constant speed mode or is controlled to maintain the cantilever deflection d constant, thus ensuring a constant force on protein, in constant force mode. Under such applied mechanical tension, the protein is stretched, transitioning into state 2 in Fig. 1C. Subsequently, one of the folded protein domains unfolds stochastically, leading to state 3, and the cantilever continues to stretch the protein to state 4. This process is repeated until all domains are unfolded or the protein link between the cantilever and the substrate is broken.

In constant speed mode, the applied force drops abruptly when a domain unfolds, resulting in a saw-tooth pattern in the force versus extension curves. Fig. 2A illustrates a representative force-extension curve for a Dys molecule with unfolding events highlighted in blue circles. Unfolding forces were recorded as a function of unfolding events, leading to statistics of force magnitude data as shown in Fig. 2B, with lines representing the kernel density estimation of the corresponding histograms. Here, a Gaussian kernel was employed to estimate the underlying probability distribution of the data collected at 1,000 nm/s of molecule Dys.

In constant force mode, unfolding events were identified by discrete steps in the molecular extension accompanied by corresponding spikes in force, as illustrated with a Dys molecule in Fig. 2C. These unfolding events are marked with blue circles and a staircase-like pattern can be observed in the extension-

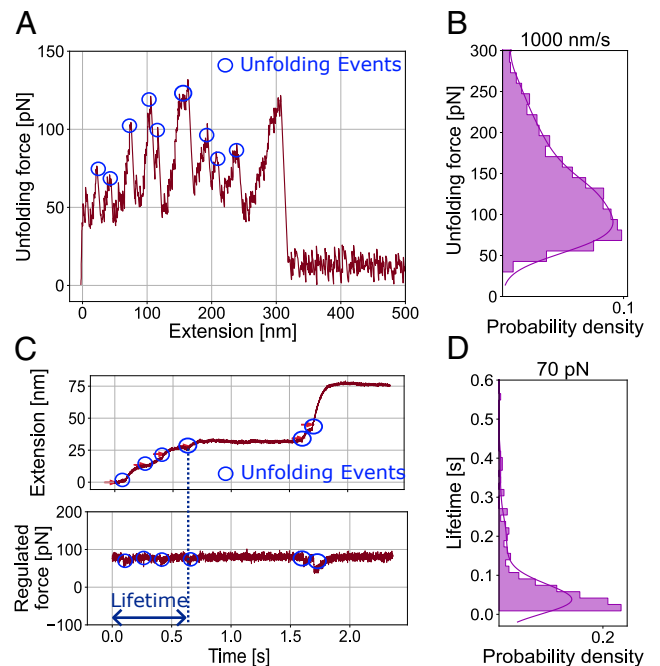


Fig. 2. SMFS using two operational modes of AFM, constant speed and constant force. (A) A representative force-extension curve of molecule Dys obtained in constant speed mode. (B) Unfolding force histograms with fitted kernel density estimation from one biological replicate of molecule Dys at 1,000 nm/s. (C) A representative curve of molecule Dys from constant force mode, with the extension-time curve in the Top panel and the force-time curve in the Bottom panel. (D) Lifetime histograms with fitted kernel density estimation from one biological replicate of molecule Dys at 70 pN. Unfolding events are highlighted with blue circles in both (A and C).

time curve. The parameter of interest is the lifetime of a bound state which is the time for an unfolding event to occur from the moment the protein is subject to tension. The lifetime histograms together with kernel density estimation is presented in Fig. 2D, where the data were collected at 70 pN of molecule Dys.

To minimize the probability of pulling multiple protein molecules, we optimized the concentration of molecules deposited (SI Appendix, Fig. S5). We set the working concentration to ensure that the probability of successful pulls remained in the 5 to 10% range (SI Appendix, Table S1 and Fig. S4), which lies within the recommended 5 to 25% window established previously (19, 20). As evidence that single molecules are being pulled, we observed that unfolded lengths increased linearly with event count, with slopes consistent with single spectrin-like repeats (SI Appendix, Fig. S13). Furthermore, when high concentrations of proteins are employed (100 nMol), transition rates $k_{off}(F)$ estimated based on constant speed and constant force data do not remain consistent (SI Appendix, Fig. S7), indicating multiple proteins are unfolded at higher concentrations. At concentrations used for the results and conclusions reached, the data from constant speed and constant force experiments exhibit consistency in transition rates $k_{off}(F)$ when analyzed using the DHS model, as illustrated in SI Appendix, Fig. S2, indicating that data primarily result from single molecule trials. Transition rates were estimated from the unfolding force distributions in the constant speed mode using SI Appendix, Eq. S3, and from lifetime averages in the constant force mode by SI Appendix, Eq. S8. In the following analysis, we focus on mean lifetimes derived from the constant force data and the most probable forces from the constant speed data, as the latter better represents the distribution than the mean.

The Mechanical Properties of Dys Differ Significantly from Utr, with Dys Exhibiting Brittle Unfolding Behavior. Data were collected across $N = 3$ preparations of both Dys and Utr using constant speed and constant force AFM modes. Violin

plots of constant speed experiments conducted at five different pulling speeds (200, 500, 1,000, 2,000, and 5,000 nm/s) are shown in Fig. 3A while the constant force data collected at five different forces (50, 60, 70, 80, and 90 pN) are presented in Fig. 3B. To further confirm that our measurements reflect single molecule events, we provide detailed histograms of the unfolding force distributions from the constant speed experiments and the lifetime distributions from the constant force experiments in SI Appendix, Figs. S8 and S9, respectively. The unfolding force distributions for Dys and Utr shift toward higher values at higher pulling speeds (Fig. 3A). To quantify the consistency across different biological replicates, we used the coefficient of variation (CV), defined as the ratio of the SD to the mean. The largest CV observed were 0.088 for Dys and 0.044 for Utr, both observed at 2,000 nm/s (SI Appendix, Table S3). The lifetime averages also exhibited a high degree of consistency among different biological replicates measured at the same regulated force and decreased with increasing regulated forces, as illustrated in Fig. 3B. The largest CV were 0.147 at 60 pN for Dys and 0.065 at 50 pN for Utr (SI Appendix, Table S3).

Notably, Dys consistently exhibited shorter lifetime averages than Utr under the same regulated forces (SI Appendix, Table S3). As the regulated force increased, the faster unfolding events reduced the resolution, narrowing the difference in lifetime averages between Utr and Dys from 0.100 s (58.8% of Dys' lifetime) at 50 pN to 0.002 s (3.8%) at 90 pN. In constant speed mode, Dys displayed lower most probable unfolding forces than Utr at same pulling speeds. The difference decreased from 15 pN (25.8% of Dys' most probable force) at 200 nm/s to 4 pN (3.3%) at 2,000 nm/s and approached zero at 5,000 nm/s. Moreover, we conducted the two sample Kolmogorov–Smirnov (KS) test (21), which tests the null hypothesis that the distributions of two samples are identical, with the level of confidence $\alpha = 0.05$. Across all pulling speeds and regulated forces, all P -values comparing Dys and Utr were below 0.05 (SI Appendix, Fig. S18), indicating significantly different

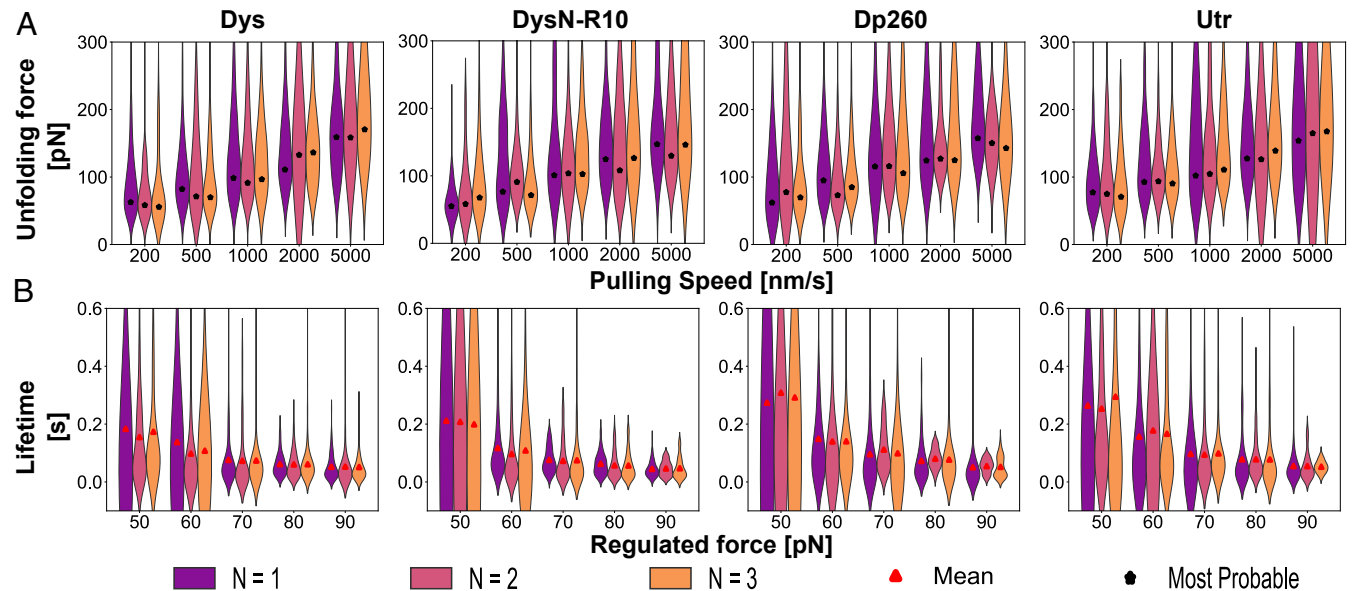


Fig. 3. Statistical summary of $N = 3$ biological replicates for (A) constant speed and (B) constant force experiments. Each column presents results for different molecules, namely Dys, DysN-R10, Dp260, and Utr from Left to Right. The data from three biological repeats are visualized in purple, pink, and yellow. Unfolding force distributions from constant speed experiments and lifetime distributions from constant force experiments are depicted using violin plots. The violin plots illustrate data distributions using kernel density estimation as black lines on each side, with the width of each curve indicating the relative frequency of data points. Meanwhile, black stars signify the positions of values with the highest probability, visualized by the widest section of the violin plot, and red triangles indicate the lifetime averages, calculated from identical forces and replicates.

mechanical properties (unfolding forces and lifetimes) between these Dys and Utr proteins.

We categorized data from all biological replicates by unfolding event counts and plotted the most probable forces from constant speed experiments of all pulling speeds, and mean dwell lifetimes from constant force measurements of all regulated forces, with their corresponding CIs (Fig. 4; see *Materials and Methods* for plotting details). Dwell lifetime, the time interval between consecutive unfolding events, accurately quantifies bond strength as additional domains become unbound.

Our previous findings (19) demonstrated that Utr exhibits a stiffening spring behavior, with most probable forces increasing as the unfolding event count (the number of domains unfolded) rises. Here, we extend this analysis to include both constant speed and constant force data, as shown in Fig. 4. The consistency of the data across these two operational modes further strengthens the interpretation established in our earlier work. For Dys, the most probable forces remained below 200 pN, and mean dwell lifetimes were consistently under 0.1 s across all unfolding event counts. When comparing Utr and Dys with a 95% CI, Utr consistently demonstrated higher unfolding forces across all events and higher mean dwell lifetimes across all events. The divergence between Dys and Utr becomes increasingly pronounced at higher unfolding event counts. For large unfolding event counts (greater than 5), error bars overlapped only at counts of 9 and 12 from constant speed data and at 6 from constant force data. To further probe these trends, we plotted histograms for Dys and Utr at early and late unfolding event counts (*SI Appendix, Fig. S15*). In both constant speed and constant force data, Dys exhibits similar distributions across early and late unfolding event count, whereas Utr shows marked differences with flatter distributions and shifted peaks under late unfolding events. These differences are also evident in additional example traces with a large number of unfolding events (*SI Appendix, Fig. S12*). Finally, we applied a substantially stricter filtering criterion to the constant speed data, which is produced in *SI Appendix, Fig. S16*, which corroborates the trend in the data shown in Fig. 4A. In summary, these findings highlight significant differences in mechanical properties between Dys and Utr, showing that Dys exhibits a brittle unfolding behavior while Utr exhibits a stiffening spring behavior.

DysN-R10 and Dp260 Exhibit Comparable Mechanical Properties to Dys, in Contrast to the Stiffening Spring Behavior of Utr. To assess whether dystrophin's brittle unfolding behavior is a property of the full-length protein or intrinsic to either the N-terminal or C-terminal half, we conducted experiments on two other dystrophin fragments, DysN-R10 and Dp260, under both constant speed and constant force modes across $N = 3$ biological replicates. The unfolding force and lifetime statistics are summarized in Fig. 3 and *SI Appendix, Table S3*. The most probable unfolding forces ($CV < 0.11$) and lifetime averages ($CV < 0.08$) were consistent across biological replicates for both DysN-R10 and Dp260. When comparing the magnitudes of most probable forces and lifetime averages across various pulling speeds and regulated forces, Dys exhibited greater or equal magnitudes than DysN-R10 in six conditions (2,000 nm/s, 5,000 nm/s, and 60 to 90 pN), and greater or equal magnitudes than Dp260 under three conditions (2,000 nm/s, 5,000 nm/s, and 90 pN). The results indicated no significant differences (P -values > 0.05) between Dys and DysN-R10 under the conditions of 200 nm/s, 1,000 nm/s, 5,000 nm/s, 80 pN, and 90 pN; and between Dys and Dp260 at 90 pN (*SI Appendix, Fig. S18*).

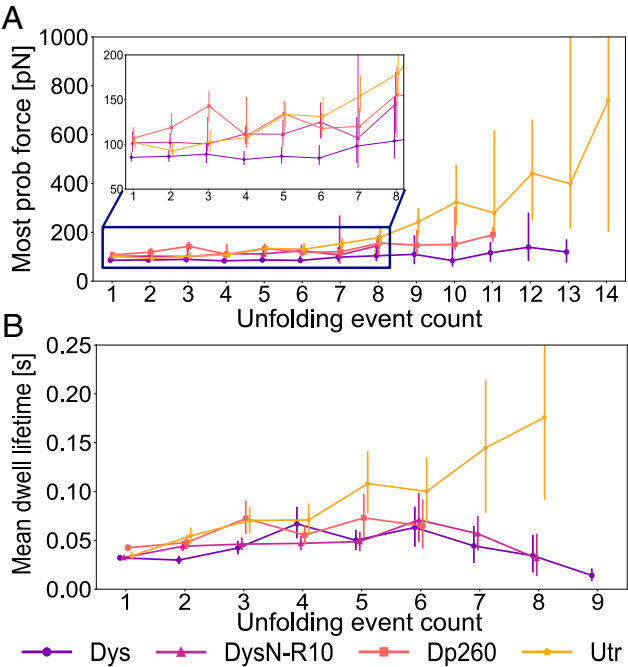


Fig. 4. Protein mechanical behavior as a function of unfolding event count (the number of domains unfolded). (A) Most probable forces from constant speed experiments. (B) Mean dwell lifetime, the time interval between consecutive unfolding events, derived from constant force experiments. Error bars represent 95% CIs, calculated using bootstrapping.

Analysis based on unfolding event counts (Fig. 4) revealed that neither DysN-R10 nor Dp260 exhibited the stiffening spring behavior observed in Utr. The most probable forces of DysN-R10 and Dp260 remained below 200 pN, and mean dwell lifetimes consistently stayed under 0.1 s across all unfolding event counts, mirroring the behavior observed in Dys. Quantitatively, the most probable force increases by $\sim 65\%$ for Dys (~ 85 pN to ~ 140 pN), by $\sim 45\%$ for DysN-R10 (~ 100 pN to ~ 145 pN), and by $\sim 90\%$ for Dp260 (~ 100 pN to ~ 190 pN), whereas Utr shows a much more pronounced increase from ~ 100 pN to over 400 pN (greater than 400% increase). In summary, the mechanical properties of DysN-R10, Dp260, and Dys are comparable, all exhibiting brittle unfolding behavior, in contrast to the stiffening spring behavior observed in Utr.

The DHS Model Reveals Similarity Among Dys Constructs and Distinction from Utr. We applied the DHS model to extract key energy landscape parameters, including the intrinsic transition rate k_0 , the barrier height $\Delta G^\ddagger$, and the distance between the folded state and the transition point along the reaction coordinate $\Delta x^\ddagger$, by fitting *SI Appendix, Eq. S4* using the least squares method. The consistency of these results was further validated by Monte Carlo simulations, as detailed in *SI Appendix, Fig. S2* for the Dys molecule; similar consistency for other molecules was summarized in *SI Appendix, Fig. S17*. Here, we present the average transition rates, k_{off} , across different operational modes and biological replicates in Fig. 5A, where the mean values and SDs are represented by lines and error bars. Additionally, corresponding energy landscapes with both cusp-like ($\nu = 1/2$) and linear cubic ($\nu = 2/3$) shapes are plotted in Fig. 5B and C, with associated parameters listed in Table 1.

Dys consistently exhibited larger transition rate averages (k_{off}) than Utr, with the difference decreasing as force increased (Fig. 5A). This observation aligns with earlier findings on com-

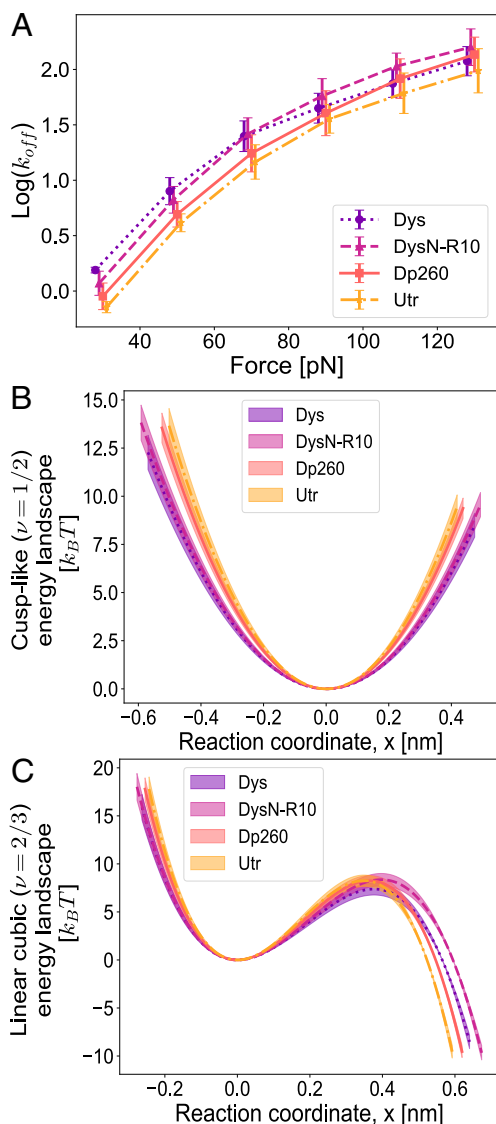


Fig. 5. Transition rates and energy landscapes of four molecules, Dys, DysN-R10, Dp260, and Utr. (A) k_{off} -vs- F curves showing mean values (lines) and SDs (error bars). (B) Cusp-like ($\nu = 1/2$) energy landscapes and (C) linear cubic ($\nu = 2/3$) energy landscapes, with mean values (lines) and SDs (shaded area). Mean and SDs are computed across different operational modes and biological replicates.

parison of lifetime averages and most probable unfolding forces (SI Appendix, Table S3). Similarly, Utr had smaller transition rate averages compared to both DysN-R10, and Dp 260. Furthermore, crossover points were observed between DysN-R10 and Dys at 70 pN, and between DysN-R10 and Dp260

at 100 pN, indicating comparable mechanical properties among these molecules.

From the perspective of energy landscape parameters, smaller $\Delta x^\ddagger$, k_0 , and larger $\Delta G^\ddagger$ correspond to a steeper energy landscape (SI Appendix, Fig. S1). We note that when k_0 are similar, $\Delta x^\ddagger$ values are generally robust to draw quantitative comparisons (22, 23). This is because the tilted energy barrier $\Delta G_F^\ddagger$ decreases faster with larger $\Delta x^\ddagger$ under external force F , following the relationship $\Delta G_F^\ddagger = \Delta G^\ddagger - F\Delta x^\ddagger$. The absolute percentage change in $\Delta x^\ddagger$ relative to Dys when compared to Utr are 11.79% and 7.44% for $\nu = 1/2$ and $\nu = 2/3$, respectively (Table 1). However, smaller percentage changes are observed when Dys is compared to DysN-R10 or Dp260. Specifically, the larger absolute percentage change in $\Delta x^\ddagger$ for $\nu = 1/2$ was 7.79% for Dys versus Dp260, and it was for 5.32% for Dys versus DysN-R10. Taken together, DysN-R10 and Dp260 exhibited more similarity to Dys when compared to Utr.

Discussion

The key conclusion from our work is that both constant speed and constant force modes of atomic force microscopy show that the mechanical properties of full-length dystrophin are distinct from utrophin. We collected a large amount of data to ensure statistically meaningful results and optimized protein concentrations to ensure that results and conclusions are based on single molecule trials (SI Appendix, Table S1). Our data from different modes exhibited consistency in transition rates k_{off} estimated from fitting the DHS model to both datasets. This conclusion is supported by Monte Carlo simulations (SI Appendix, Fig. S17). At higher concentrations, however, this consistency breaks down (SI Appendix, Fig. S7), further supporting that our measurements reflect single-molecule events rather than simultaneous pulling of multiple proteins.

Previous SMFS studies showed that small dystrophin fragments had unfolding forces of ~ 20 to 50 pN (10, 24, 25), while we demonstrated that full-length utrophin and large utrophin fragments, comprising approximately half the molecule, unfold with forces of ~ 100 pN (19). Our subsequent study showed that differences in protein expression system and phosphorylation partially account for the differences in unfolding forces between our study of utrophin and prior studies of dystrophin fragments (20). Therefore, here we directly compared the mechanical properties of recombinant utrophin and dystrophin expressed using the same cell expression and AFM systems.

Our data showed that Dys exhibits uniform unfolding properties also displayed by its N- and C-terminal large fragments, DysN-R10 and Dp260 (Fig. 3 and SI Appendix, Fig. S18 and Table S3). All three molecules displayed brittle unfolding behavior, with both the most probable unfolding forces and

Table 1. DHS model parameters of four molecules: Dys, DysN-R10, Dp260, and Utr

Molecules	DHS $\nu = 1/2$				DHS $\nu = 2/3$			
	$\ln(k_0)$	$\Delta x^\ddagger$ [nm]	$\Delta G^\ddagger$ [$k_B T$]	$\Delta x^\ddagger$ diff (%)	$\ln(k_0)$	$\Delta x^\ddagger$ [nm]	$\Delta G^\ddagger$ [$k_B T$]	$\Delta x^\ddagger$ diff (%)
Dys	-2.363 ± 0.243	0.475 ± 0.028	8.528 ± 0.620	0	-1.809 ± 0.319	0.376 ± 0.026	7.376 ± 0.617	0
DysN-R10	-2.795 ± 0.453	0.493 ± 0.052	9.599 ± 0.633	3.79	-2.209 ± 0.417	0.396 ± 0.040	8.361 ± 0.643	5.32
Dp 260	-2.678 ± 0.206	0.438 ± 0.018	9.408 ± 0.536	7.79	-2.242 ± 0.280	0.364 ± 0.020	8.250 ± 0.542	3.19
Utr	-2.719 ± 0.344	0.419 ± 0.040	9.489 ± 0.622	11.79	-2.287 ± 0.294	0.348 ± 0.025	8.256 ± 0.582	7.44

^aEnergy landscapes depicted in Fig. 5 are based on the parameters provided here.

^b $\Delta x^\ddagger$ difference represents the absolute percentage change in $\Delta x^\ddagger$ relative to Dys.

lifetime averages remaining consistent as the unfolding event count increases. Previous studies showed that different fragments of dystrophin's central rod domain exhibited similar unfolding forces (10) and that forces to unfold a minidystrophin appeared the same as its shorter core construct (24). These observations together with our results of longer dystrophin constructs further corroborate a brittle unfolding behavior for dystrophin. The unfolding behavior of dystrophin and its large fragments contrasts strikingly with the stiffening spring behavior that we previously reported for utrophin (19) and confirmed here (Fig. 4). Notably, these divergent mechanical behaviors become most apparent at higher unfolding event counts, as also observed in our previous findings (19), where the differences between utrophin and dystrophin are amplified and not explained by the properties of smaller fragments alone. Furthermore, energy landscapes supported these findings (Fig. 5 and Table 1), showing that Utr is positioned significantly further away from Dys compared to DysN-R10 and Dp260.

Our study quantifies the mechanical properties of dystrophin, its large fragments (DysN-R10 and Dp260), and utrophin using both constant speed and constant force atomic force microscopy. Data from different modes, with consistency via the DHS model, revealed two distinct mechanical behaviors. Dystrophin and its fragments exhibit a brittle unfolding behavior, where sequentially unfolding domains display similar mechanical properties. In contrast, utrophin is a stiffening spring, with increasing resistance to unfolding events. While these single-molecule measurements capture the mechanical properties of isolated molecules, their behaviors in living cells are likely influenced by interactions with actin, the plasma membrane, and associated binding proteins such as syntrophin, as well as by posttranslational modifications. These cellular factors may modulate their mechanical responses under physiological conditions, and future studies integrating single molecule and cellular approaches will be important for fully elucidating their functional roles. Although AFM-based force spectroscopy probes forces higher than those expected in vivo, it offers valuable insight about how these proteins may behave. Notably, the mechanical differences between utrophin and dystrophin are increasingly pronounced at lower pulling speeds and regulated forces, suggesting that in the physiological regime of a few piconewtons, the contrast is likely to be even more significant.

We hypothesize that dystrophin's brittle unfolding behavior allows it to dissipate mechanical energy during muscle stretching through stochastic domain unfolding, effectively protecting the sarcolemma from mechanical tearing or rupture. In contrast, utrophin's stiffening spring behavior concentrates rather than dissipates mechanical stress, potentially exacerbating membrane damage through sudden energy release when forces are abruptly removed. These contrasting behaviors reinforce the hypothesis that dystrophin functions as a molecular shock absorber, protecting the sarcolemma from mechanical stress.

Utrophin's stiffening behavior, however, appears well-suited to its native localization at the myotendinous junction (14), where its primary role involves efficient mechanical force transfer rather than energy dissipation. This functional specialization is further supported by in vivo evidence showing decreased passive stiffness measured at the tendons of skeletal muscles isolated from utrophin knockout mice (19). Thus, utrophin's mechanical behavior at the level of single molecules is consistent with its role as a stiff elastic element working in series with titin at myotendinous junctions. In contrast, the brittle unfolding behavior of dystrophin seems consistent with its hypothesized function

as a membrane shock absorber. Interestingly, replacement of 4 dystrophin repeats with 4 α -actinin repeats in a microdystrophin transgene construct failed to rescue the dystrophic phenotypes of mdx mice (26). While transgenic overexpression of full-length utrophin effectively rescues most mdx phenotypes (15), it does not protect mdx muscle from eccentric contraction-induced force loss in vivo as effectively as dystrophin or minidystrophin transgenes (27). Taken together, these findings suggest that dystrophin and utrophin perform different mechanical functions in muscle cells, and that utrophin may not be an ideal therapeutic surrogate for dystrophin deficiency in DMD because it exhibits a stiffening spring mechanical response compared to the brittle shock absorber behavior exhibited by dystrophin.

Materials and Methods

Cloning. Dys, DysN-R10, Dp260, and Utr were cloned as previously described (28). All constructs contained an N-terminal FLAG-tag for affinity purification (19, 28).

Protein Expression and Purification. Insect cells used for protein expression were maintained at 1×10^6 cells/mL in 250 mL suspension cultures. Sf9 cells and Tni cells were grown in Sf-900™ II SFM (Thermo Fisher Scientific) and ESF 921™ (Expression Systems), respectively, both supplemented with penicillin/streptomycin (Sigma-Aldrich) and fungizone (Thermo Fisher Scientific) in a 250 mL suspension culture. The baculovirus generation for Utr was previously described (20). For Dys, DysN-R10, and Dp260, higher-titer viral stocks were generated through successive infections of Sf9 cells in 6 cm (P0), 10 cm (P1), and 15 cm (P2) plates. For P0, 2.5×10^6 cells were plated and allowed to adhere for 1 h before transfection with a bacmid and Cellfectin mix. After 5 h at 27 °C, extant media were removed, 4 mL Grace's complete media (Thermo Fisher Scientific), FBS, antibiotics, and antifungal were added, and the plates were incubated at 27 °C for 4 d. For P1, 10×10^6 cells were plated and incubated for 1 h at room temperature before adding 1 mL of P0 virus to each plate. The cells were then incubated for another hour at room temperature, with gentle tilting every 15 min to ensure even virus distribution. Afterward, 9 mL Grace's complete media were added, and the plates were incubated at 27 °C for 3 d. For P2, 22×10^6 cells were plated on a 15 cm plate and incubated for 1 h at room temperature before adding 0.5 mL of P1 virus. The cells were incubated for another hour, with gentle tilting every 15 min to ensure uniform virus distribution. 15 mL Grace's complete media were then added, and the plates were incubated at 27 °C for 3 d. Dys were expressed in Tni cells, Dp260 and DysN-R10 were expressed in Sf9 cells, and Utr were expressed in both Tni and Sf9 cells.

For protein expression, 250 mL 1×10^6 cells/mL cultures were transfected with 10 mL of conditioned media baculovirus and incubated at 140 rpm for 72 h at 27 °C. The cells were harvested by centrifuging at $500 \times g$ for 5 min at 4 °C. Cell pellets were resuspended in I-PER lysis buffer (Thermo Fisher Scientific), vortexed on medium for 5 s, then incubated on ice for 10 min. Each lysate was then centrifuged at $15,000 \times g$ for 15 min at 4 °C to clear the lysates. After centrifugation, EDTA was added to the supernatant to reach a final concentration of 0.2 mM. The supernatant was applied to an anti-FLAG M2 agarose column (Sigma-Aldrich), washed with 30 mL of molecular-grade phosphate-buffered saline (PBS), and eluted with 100 μ g/mL FLAG peptide in PBS. The I-PER lysis buffer and PBS used for wash and elution had 100 nM aprotinin, 1 μ g/mL pepstatin A, 10 μ g/mL E-64, 10 μ M leupeptin, 1 mM PMSF, and 2 mM benzamidin. For Dp260, DysN-R10, and Utr, 5 mL of the PBS + FLAG peptide (elution buffer) was applied to the resin, with 3 mL drained and collected in 1 mL fractions. The remaining 2 mL was left on top of the resin for 1 h. Another 5 mL of elution buffer was added to the resin, drained, and collected in 1 mL fractions. The fourth fraction of elution was used for atomic force microscopy experiments. For Dys, 5 mL of the PBS + FLAG peptide (elution buffer) was applied to the resin, with 3 mL drained and collected in 1×3 mL, 3×1 mL, or 6×0.5 mL fractions. The remaining 2 mL was left on top of the

resin for 1 h. Another 5 mL of elution buffer (or 25 mL in the case of 1×3 mL fractions) was added to the resin, drained, and collected in 3 mL, 1 mL, or 0.5 mL fractions, respectively. The second, fourth, or eighth fraction (respectively) of elution was used for atomic force microscopy experiments. The eluted protein was dialyzed for 1 h in 1 L of 20 mM HEPES (pH 7.5), 150 mM NaCl, 1 mM DTT, followed by overnight dialysis in another liter of buffer with the same composition. Protein concentration was determined by Bradford assay using an Albumin standard curve. Purified proteins were run on a 3 to 12% sodium dodecyl sulfate (SDS) polyacrylamide gradient gel and stained with Coomassie blue stain for verification (Fig. 1B). Coomassie stained gels were visualized and imaged using Licor Odyssey® Infrared Imaging System. Sucrose was added to unused protein elutions to reach a final concentration of 150 mM, and the protein elutions were then flash-frozen in liquid nitrogen.

Atomic Force Microscopy Experiments. Single molecule force experiments were conducted using MFP-3D atomic force microscope (AFM) (Asylum Research, an Oxford Instruments Company, Santa Barbara, CA). The AFM configuration consists of a flexible cantilever with a sharp tip, a laser-photodiode based sensor for tracking the cantilever tip's position, and a piezoelectric nanopositioner capable of moving the substrate in three dimensions relative to the cantilever base (29). We employed Bruker MLCT-BIO triangular cantilevers, made of silicon nitride and coated with reflective gold on the backside, with a nominal spring constant of 10 pN/nm and a typical tip radius of 20 nm. Prior to each experiment, the spring constant was determined by analyzing the thermally excited deflection of the cantilever (30). A droplet (~100 μ l) of purified protein suspended in PBS, with a dilution ranging from 3 to 30 nanomolar concentration (nMol), was placed on a freshly cleaved and ionized mica substrate. The protein concentrations were chosen to ensure a 5 to 10% probability of successful pulls, a success percentage used in previous studies to minimize the probability of pulling multiple proteins (19, 31). This setup was incubated for 15 min to allow the protein to adhere to the mica surface. The droplet containing the protein solution was then removed and the mica substrate was washed once with 100 μ l of PBS to eliminate proteins that had not adhered. A new droplet containing 100 μ l of PBS was introduced to establish the environment for the force spectroscopy experiments. Repeated approach-retraction cycles were conducted at room temperature (21 to 23 °C), where the cantilever's tip was pressed against the substrate for a duration of 2 s, applying an indentation force of 600 pN. Subsequently, in the constant speed mode, the cantilever base was retracted at a speed ranging from 200 to 5,000 nm/s. The pulling force on the molecule is balanced by the force experienced by the cantilever, thus the force, F on the protein was determined as,

$$F = k_c d, \quad [1]$$

where k_c is the spring constant of the cantilever and d is the deflection of the cantilever. In constant force mode, the cantilever base was regulated to maintain a steady deflection, resulting in a regulated force on the molecule, ranging from 50 to 90 pN. Data from 300 to 2,000 successful force spectroscopy experiments (with at least one identifiable unfolding event) were collected for each protein under a specific speed or force condition and used to determine the unfolding statistics.

Data Analysis. The raw data obtained from the force spectroscopy experiments are primarily composed of three main variables: 1) the time for a bond to unravel and how the forces vary with time, 2) the deflection of the cantilever, and 3) the distance between the base of the cantilever and the substrate. It includes instances where there is no protein between the tip and substrate, as well as

instances where a protein was effectively adsorbed and subsequently stretched. To identify successful experiments and extract unfolding events from data, we utilize a custom algorithm developed in MATLAB R2023b (MathWorks) (19, 20). The algorithm calculates the applied force by multiplying the cantilever deflection with the spring constant and computes the protein extension by subtracting the deflection of the cantilever from the distance between the cantilever base and the substrate. The algorithm first computes the SD σ of noise in the force signal measured from the approach curve. Subsequently, in constant speed mode, it detects characteristic peaks in the force signal that exceed 3σ . In constant force mode, it identifies steps larger than 10σ in the extension signal coinciding with a spike in the force signal. Here, the significant events are the peaks in constant speed mode and steps in constant force mode. The first significant event arises from the adhesive force between the cantilever tip and the substrate; this event is excluded from the dataset for subsequent analysis. Similarly, the last significant event, often affected by a detachment event, is also discarded. The remaining significant events are recognized as domain unfolding events, which are of interest. The number of significant events is then used to distinguish between successful and unsuccessful experiments.

The force-extension curves obtained from the constant speed mode are fitted to a worm-like chain (WLC) model. The WLC model (32) relates the force applied on protein F_{WLC} to its extension q , and is given by,

$$F_{WLC}(q, L_c, L_p) = \frac{k_B T}{L_p} \left[\frac{1}{4(1 - \frac{q}{L_c})^2} - \frac{1}{4} + \frac{q}{L_c} \right], \quad [2]$$

where k_B is the Boltzmann constant, T is temperature, and L_c and L_p are the contour length and the persistence length of the protein, respectively. For the constant force mode, extension-time curves are fitted to a staircase function. These curves are then inspected manually. Curves that show irregularities, such as an excessive number of unfolding events or distorted significant events that deviate from the characteristics of the fitting curve, are excluded for further analysis.

To analyze mechanical properties across different unfolding event counts, we first categorized data from all biological replicates, retaining only categories with sample sizes greater than 10 to minimize bias. After removing outliers (greater than three SDs from the mean), we performed bootstrapping by resampling with replacement to generate estimate distributions. This approach allowed us to determine CIs using percentile methods for both: 1) the most probable unfolding forces from constant speed measurements and 2) the mean dwell lifetimes from constant force experiments.

Calibration Protein. Titin I270 (Athena Enzyme Systems™), an AFM reference protein composed of eight repeats of the Ig 27 domain of human titin, serves as the calibration protein. The most probable unfolding force at a pulling speed of 1,000 nm/s is 219.40 pN (SI Appendix, Fig. S3C), closely matching the reported value of 225 pN (33). Additionally, energy landscape parameters align with values previously reported in the literature (34), where the literature values are represented by dashed lines in SI Appendix, Fig. S3 F and G.

Data, Materials, and Software Availability. The data presented in the article are available via Dryad (35). All other data are included in the article and/or SI Appendix.

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